BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The brainstem

The written notes for the brainstem lecture: the three regions in order, the midbrain, pons, and medulla structures, the cranial nerve nuclei by region, and the reticular formation. Module 5 competencies 161 to 166, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

> **How to use these notes.** Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 161. Brainstem regions

In your course notes: Module 5 packet, Chapter 5, competency 161 (page 42).

Dr. Rennie's video for this part: The brainstem concept video.

You should be able to: Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

The brainstem is the stalk of the brain. It sits below the diencephalon, in front of the cerebellum, and above the spinal cord. It has three regions. From top to bottom they are the midbrain, the pons, and the medulla oblongata.

The three regions, top to bottom

Midbrain
: Also called the mesencephalon. The short, narrow region just below the diencephalon. It is the most superior region of the brainstem.

Pons
: The rounded bulge below the midbrain. Its name means bridge. On the model, the rounded bulge on the front of the pons is the basilar pons.

Medulla oblongata
: The lowest region. It tapers down and becomes the spinal cord at the foramen magnum. The medulla, not the pons, is the region that turns into the spinal cord.

On the front of a brain model or sheep brain, look for the groove where the pons meets the medulla. This is the pontomedullary junction. It is a useful landmark, because three cranial nerves (VI, VII, and VIII) leave the brainstem along it.

Region	Where it is	What it connects
Midbrain	Most superior region; just below the diencephalon	Joins the diencephalon to the pons; the superior cerebellar peduncles attach here
Pons	Middle region; the rounded bulge in front of the cerebellum	Bridges the midbrain and the medulla, and links the cerebrum to the cerebellum through the middle cerebellar peduncles
Medulla oblongata	Most inferior region; ends at the foramen magnum	Continuous with the spinal cord below; the inferior cerebellar peduncles attach here
Pontomedullary junction	The groove between the pons and the medulla on the front surface	Marks the border of the two regions; CN VI, VII, and VIII leave here

How the brainstem connects the higher brain to the spinal cord

1. Every tract passes through it. Every ascending tract carrying sensory signals up to the brain and every descending tract carrying motor commands down to the cord runs through all three regions.
2. It holds cranial nerve nuclei. The nuclei of cranial nerves III through X and XII are in the brainstem, so ten of the twelve cranial nerves have brainstem nuclei. The nucleus of XI sits just below it, in the upper cervical spinal cord.
3. It carries the cerebellum. Three pairs of cerebellar peduncles tie the cerebellum to it: the superior peduncles to the midbrain, the middle peduncles to the pons, and the inferior peduncles to the medulla.

- Memory tip: read the brainstem from the top down: Midbrain, then Pons, then Medulla (M, P, M). The medulla is the one that turns into the spinal cord.
- One structure you can find on each region on the model: the cerebral peduncles or cerebral aqueduct on the midbrain, the basilar pons or middle cerebellar peduncle on the pons, and the pyramids or medullary olive on the medulla.

Competency 162. Midbrain structures

In your course notes: Module 5 packet, Chapter 5, competency 162 (page 43).

Dr. Rennie's video for this part: The brainstem concept video.

You should be able to: Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

The midbrain surrounds a narrow channel, the cerebral aqueduct. The aqueduct carries cerebrospinal fluid from the third ventricle above down to the fourth ventricle below. On a cross section of the midbrain it is the small hole near the back. Use it to orient the specimen: everything behind the aqueduct is the roof, and everything in front of it is the floor and the stalks.

The roof: the tectum and its four colliculi

The roof of the midbrain is the tectum. Seen from behind, it carries four rounded bumps, together called the corpora quadrigemina. The upper pair are the superior colliculi. They are reflex centers for vision, such as turning the eyes and head toward a moving object.

The lower pair are the inferior colliculi. They are reflex centers and relay stations for hearing, such as turning the head toward a sudden sound. You do not become conscious of the sight or sound in the colliculi; they run reflexes only. Awareness of what you see and hear happens in the cerebral cortex.

The front: the cerebral peduncles, the crus cerebri, and the substantia nigra

The front of the midbrain is formed by the two cerebral peduncles. These thick stalks carry the descending motor tracts from the cerebrum down to the pons, medulla, and spinal cord. The most anterior part of each peduncle is a band of white matter called the crus cerebri. Between the aqueduct and the crura lies the tegmentum, the core of the midbrain, which holds cranial nerve nuclei and part of the reticular formation.

The substantia nigra is a dark band of gray matter between the crus cerebri and the tegmentum, so on a cross section it sits just behind the crus cerebri. Its neurons contain a dark pigment, which is why it looks black in a fresh cross section. It works with the basal nuclei (basal ganglia) to help control movement.

Two cranial nerves leave the midbrain. The oculomotor nerve (CN III) leaves the front, in the gap between the two cerebral peduncles. The trochlear nerve (CN IV) leaves the back, just below the inferior colliculi.

Structure	Where it is	What it does
Tectum	The roof of the midbrain, behind the cerebral aqueduct	Carries the four colliculi
Superior colliculi	The upper pair of bumps on the tectum	Reflex centers for vision, such as tracking a moving object
Inferior colliculi	The lower pair of bumps on the tectum	Reflex centers and relays for hearing, such as turning toward a sound
Cerebral peduncles	The paired stalks on the front of the midbrain	Carry descending motor tracts from the cerebrum
Crus cerebri	The most anterior part of each cerebral peduncle	The white matter band holding the descending motor fibers
Substantia nigra	A dark band between the crus cerebri and the tegmentum	Works with the basal nuclei to help control movement
Cerebral aqueduct	The narrow channel through the midbrain, between the tectum and the tegmentum	Carries CSF from the third ventricle to the fourth ventricle

- Memory tip: superior colliculi sit higher and serve sight; inferior colliculi sit lower and serve sound. Both are reflex centers. You become aware of what you see and hear in the cerebral cortex.
- Cerebrospinal fluid runs downward through the aqueduct: in from the third ventricle above, out to the fourth ventricle below.

Competency 163. Pons structures

In your course notes: Module 5 packet, Chapter 5, competency 163 (page 44).

Dr. Rennie's video for this part: The brainstem concept video.

You should be able to: Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

The pons is the rounded bulge between the midbrain and the medulla. The word pons means bridge, and it bridges three regions: the midbrain above it, the medulla below it, and the cerebellum behind it. Its front surface is the basilar pons, a thick band of fibers that runs crosswise and curves back on each side toward the cerebellum. The back part of the pons is its tegmentum, which carries cranial nerve nuclei and part of the reticular formation.

The pontine nuclei and the middle cerebellar peduncles

Scattered through the basilar pons are the pontine nuclei. These clusters of gray matter receive motor information from the cerebral cortex. Their axons cross to the opposite side and enter the cerebellum. This relay is how the cerebrum keeps the cerebellum informed about planned movements, so the cerebellum can coordinate them.

The axons from the pontine nuclei leave the pons as the middle cerebellar peduncles. These are the largest of the three pairs of cerebellar peduncles. They are the thick stalks that attach the side of the pons to each cerebellar hemisphere. On a model, follow the crosswise fibers of the basilar pons around the side and into the cerebellum. Signals from the pons enter the cerebellum through the middle cerebellar peduncle, not the superior or inferior one.

Motor information from the cortex to the cerebellum, in order

1. Cerebral cortex: sends the motor information.
2. Pontine nuclei: relay the motor information on to the cerebellum.
3. Middle cerebellar peduncles: carry the signals from the pontine nuclei into the cerebellum.

4. Cerebellum: uses the information to coordinate movement.

The fourth ventricle

Behind the pons lies the fourth ventricle. It is a diamond-shaped space filled with cerebrospinal fluid. Its floor is the back of the pons and the upper medulla, and the cerebellum lies over its roof. So the fourth ventricle sits between the brainstem and the cerebellum; it is not inside the pons. The cerebral aqueduct opens into it from above.

The pons also holds nuclei that help adjust the breathing rhythm set by the medulla. The trigeminal nerve (CN V) leaves the side of the pons. The abducens (CN VI), facial (CN VII), and vestibulocochlear (CN VIII) nerves leave at the pontomedullary junction.

Structure	Where it is	What it does
Basilar pons	The front surface of the pons	Crosswise fibers heading to the cerebellum, with the pontine nuclei scattered among them
Pontine nuclei	Clusters of gray matter in the basilar pons	Relay motor information from the cerebral cortex to the cerebellum
Middle cerebellar peduncles	The thick stalks on each side of the pons	Carry the axons of the pontine nuclei into the cerebellum
Fourth ventricle	Behind the pons and upper medulla, in front of the cerebellum	Holds CSF; receives it from the cerebral aqueduct

Competency 164. Medulla oblongata structures

In your course notes: Module 5 packet, Chapter 5, competency 164 (page 45).

Dr. Rennie's video for this part: The brainstem concept video.

You should be able to: Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

The medulla oblongata is the lowest region of the brainstem. It runs from the pontomedullary junction down to the foramen magnum, where it becomes the spinal cord. Its upper half forms the lower part of the floor of the fourth ventricle. Its lower half surrounds a narrow channel that continues into the cord as the central canal.

On the surface: pyramids, decussation, and olives

Pyramids

The front of the medulla has two long ridges, one on each side of the midline. Each pyramid is made of the corticospinal tract, the descending motor pathway from the cerebral cortex to the spinal cord.

Decussation of the pyramids

Near the bottom of the medulla, just above the spinal cord, most corticospinal fibers cross the midline to the opposite side. Decussation means crossing. On a model it shows as fibers that cross over and blur the groove between the two pyramids. Because of this crossing, the left cerebral hemisphere controls the muscles on the right side of the body, and the right hemisphere controls the left side.

Olive (the medullary olive)

Just lateral to each pyramid is an oval swelling, the olive. The olives sit to the outside of the pyramids, not between them. Each is formed by the inferior olivary nucleus underneath, which relays information to the cerebellum through the inferior cerebellar peduncle.

On the front of the medulla, the hypoglossal nerve (CN XII) leaves in the groove between the pyramid and the olive. The glossopharyngeal (CN IX) and vagus (CN X) nerves leave in the groove behind the olive.

Inside: the three vital centers

The medulla also holds three vital centers in its core. They are part of the reticular formation and are not visible on the surface. The cardiac center adjusts the rate and force of the heartbeat. The respiratory center sets the basic rhythm of breathing. The vasomotor

center adjusts the diameter of blood vessels and so helps set blood pressure. Only the pyramids, olives, and decussation show on the surface; the three centers are inside.

Structure	Where it is	What it is or does
Pyramids	Paired ridges on the front of the medulla (surface)	Formed by the descending corticospinal tracts
Decussation of the pyramids	The lower medulla, just above the spinal cord (surface)	Where most corticospinal fibers cross to the opposite side
Olive	Oval bulge lateral to each pyramid (surface)	Formed by the inferior olivary nucleus, a relay to the cerebellum
Cardiac center	Inside the medulla (not visible on the surface)	Adjusts the rate and force of the heartbeat
Respiratory center	Inside the medulla (not visible on the surface)	Sets the basic rhythm of breathing
Vasomotor center	Inside the medulla (not visible on the surface)	Adjusts blood vessel diameter, and so blood pressure

What this means for patients

- The medulla oblongata holds the cardiac center, respiratory center, and vasomotor center. Because these centers control the heartbeat and breathing, an injury to the medulla can be life-threatening right away, in a way that an injury to a single area of cerebral cortex is not.
- The decussation of the pyramids is the anatomical reason a stroke in the right cerebral hemisphere weakens the left side of the body. Most of the motor fibers have already crossed by the time they reach the spinal cord.
- The reticular activating system links the brainstem to consciousness. Damage to this ascending network in the upper brainstem can cause a coma, even when the cerebral cortex itself is not injured.
- The substantia nigra links the brainstem to movement: loss of its pigmented neurons is the basis of Parkinson disease.

Competency 165. Cranial nerve nuclei distribution

In your course notes: Module 5 packet, Chapter 5, competency 165 (page 46).

Dr. Rennie's video for this part: The brainstem concept video.

You should be able to: Match cranial nerve nuclei III through XII to the brainstem region that houses them.

A cranial nerve nucleus is a cluster of neuron cell bodies inside the brain that gives rise to the motor fibers of a cranial nerve or receives its sensory fibers. Cranial nerves I (olfactory) and II (optic) connect to the forebrain, so they have no brainstem nucleus. The nuclei of cranial nerves III through X and XII are housed in the brainstem, and the nucleus of XI sits just below it in the upper cervical spinal cord. That makes 10 of the 12 cranial nerves with brainstem nuclei.

The nuclei line up from top to bottom in the same order as the nerve numbers. The midbrain holds III and IV. The pons holds V, VI, VII, and VIII. The medulla holds IX, X, and XII. A nerve usually leaves the brainstem near the level of its nucleus, so the attachment point on a model tells you the region.

Three nerves that need a closer look

Trigeminal (CN V)

Assigned to the pons, where its main motor and sensory nuclei sit, but its sensory nuclei stretch up into the midbrain and down through the medulla.

Vestibulocochlear (CN VIII)

Its vestibular and cochlear nuclei sit at the junction of the pons and medulla, so it is listed with both. It is the one nerve that belongs beside two regions.

Accessory (CN XI)

Grouped with the medulla in the 2-2-4-4 rule below, but its motor nucleus sits in the upper cervical spinal cord. Its rootlets rise through the foramen magnum and then leave the skull with the vagus nerve.

Cranial nerve	Region that houses the nucleus	Where the nerve leaves the brainstem
III Oculomotor	Midbrain	Front of the midbrain, between the cerebral peduncles
IV Trochlear	Midbrain	Back of the midbrain, just below the inferior colliculi
V Trigeminal	Pons (sensory nuclei extend into the midbrain and medulla)	Side of the pons
VI Abducens	Pons	Pontomedullary junction, near the midline
VII Facial	Pons	Pontomedullary junction, lateral to VI
VIII Vestibulocochlear	Pons and medulla, at their junction	Pontomedullary junction, lateral to VII
IX Glossopharyngeal	Medulla	Groove behind the olive
X Vagus	Medulla	Groove behind the olive, below IX
XI Accessory	Upper cervical spinal cord (grouped with the medulla)	Side of the upper cervical cord; rootlets rise through the foramen magnum
XII Hypoglossal	Medulla	Groove between the pyramid and the olive

- Memory tip: use the 2-2-4-4 rule. Two nerves (I and II) attach above the brainstem, two (III and IV) to the midbrain, four (V to VIII) to the pons, and four (IX to XII) to the medulla. Remember that VIII also reaches the medulla and that XI starts in the cervical cord.
- Abducens (VI) and facial (VII) belong to the pons, not the medulla. They leave at the pontomedullary junction, which is the border, but their nuclei are on the pons side of it.

Competency 166. Reticular formation

In your course notes: Module 5 packet, Chapter 5, competency 166 (page 47).

Dr. Rennie's video for this part: The brainstem concept video.

You should be able to: Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

The reticular formation is a loose net of gray matter that runs through the core of the whole brainstem, from the medulla up through the pons and the midbrain. Its name comes from the Latin for net. It is made of small clusters of neurons scattered among bundles of white matter, so it has no sharp borders and no surface landmark. It runs through all three brainstem regions, not only the medulla.

To locate it, look at any brainstem cross section. The reticular formation fills the central core (the tegmentum), behind the large descending motor tracts and in front of the ventricle or aqueduct. Its neurons connect with the cranial nerve nuclei, the cerebellum, the thalamus, and the spinal cord.

Two components running in opposite directions

The ascending reticular activating system, or RAS, sends fibers up through the thalamus to the cerebral cortex. It keeps the cortex awake and alert, holds attention, and filters out unimportant sensory input so the cortex is not flooded. RAS activity falls during sleep: when the RAS shuts down, the result is sleep. Severe damage to it can cause a coma.

The descending component sends fibers down to the spinal cord, and through it acts on the skeletal muscles.

These fibers help set the resting tone of skeletal muscles and help keep posture. The cardiac, respiratory, and vasomotor centers of the medulla are also part of the reticular formation, so this descending output links the brainstem to heart rate, breathing, and blood vessel diameter.

Component	Direction and target	What it does
Ascending reticular activating system (RAS)	Up through the thalamus to the cerebral cortex	Keeps the cortex awake and alert (consciousness and arousal), holds attention, filters out unimportant sensory input
Descending component	Down to the spinal cord, and through it the skeletal muscles	Helps set resting muscle tone and posture; includes the medullary centers for heart rate, breathing, and blood vessel diameter

- Memory tip: ascending means awake: the ascending system wakes up the cortex. Descending means down to the muscles: it sets their tone.
- Muscle tone belongs to the descending component. The RAS is the upward part that keeps the cortex awake.

Worked questions

Eight questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 161

Which region of the brainstem becomes the spinal cord at the foramen magnum?

A. Midbrain

B. Pons

C. Medulla oblongata

D. Diencephalon

Answer: C. The medulla oblongata is the most inferior region of the brainstem. It tapers down and is continuous with the spinal cord at the foramen magnum. Read the brainstem top to bottom: midbrain, pons, medulla.

Why the others are wrong

- **A** The midbrain is the most superior region, just below the diencephalon. It joins the diencephalon to the pons.
- **B** The pons is the middle region, the rounded bulge in front of the cerebellum. It bridges the midbrain and the medulla, but it does not reach the spinal cord.
- **D** The diencephalon is not part of the brainstem at all. It sits above the midbrain, between the cerebrum and the brainstem.

QUESTION 2 · COMPETENCY 162

The superior colliculi of the tectum are reflex centers for

A. vision, such as turning the eyes toward a moving object

B. hearing, such as turning the head toward a sound

C. voluntary movement, working with the basal nuclei

D. moving cerebrospinal fluid from the third to the fourth ventricle

Answer: A. The superior colliculi are the upper pair of bumps on the tectum and they run visual reflexes. Superior sits higher and serves sight. You do not become conscious of the sight there; awareness happens in the cerebral cortex.

Why the others are wrong

- **B** Hearing reflexes belong to the inferior colliculi, the lower pair of bumps on the tectum.
- **C** Helping control movement with the basal nuclei is the job of the substantia nigra, the dark band behind the crus cerebri.
- **D** Cerebrospinal fluid moves through the cerebral aqueduct, the narrow channel in front of the tectum. The colliculi have nothing to do with CSF.

QUESTION 3 · COMPETENCY 163

Axons from the pontine nuclei enter the cerebellum through which structure?

A. Superior cerebellar peduncles

B. Inferior cerebellar peduncles

C. Cerebral peduncles

D. Middle cerebellar peduncles

Answer: D. The pontine nuclei relay motor information from the cerebral cortex to the cerebellum, and their axons leave the pons as the middle cerebellar peduncles, the thick stalks on each side of the pons and the largest of the three pairs.

Why the others are wrong

- **A** The superior cerebellar peduncles attach the cerebellum to the midbrain, not the pons.
- **B** The inferior cerebellar peduncles attach the cerebellum to the medulla. The inferior olivary nucleus relays through them.
- **C** The cerebral peduncles are the stalks on the front of the midbrain that carry descending motor tracts from the cerebrum. They are not cerebellar peduncles.

QUESTION 4 · COMPETENCY 164

A stroke in the right cerebral hemisphere weakens the left side of the body. Which structure explains this?

A. The olives, which relay motor commands to the opposite cerebellar hemisphere

B. The decussation of the pyramids, where most corticospinal fibers cross to the opposite side

C. The pyramids, which carry fibers from the right hemisphere only

D. The pontine nuclei, which send axons to the opposite side of the cerebellum

Answer: B. The corticospinal tract runs down through the pyramids, and near the bottom of the medulla most of its fibers cross the midline at the decussation of the pyramids. So the right hemisphere ends up controlling the muscles on the left side of the body.

Why the others are wrong

- **A** The olives are formed by the inferior olivary nucleus, which relays information to the cerebellum. They do not carry the corticospinal tract.
- **C** There are two pyramids, one on each side of the midline, and each carries the corticospinal tract from its own hemisphere. The crossing happens below them, at the decussation.
- **D** The pontine nuclei do send crossed axons into the cerebellum, but that pathway informs the cerebellum about planned movements. It is not the pathway that moves the body.

QUESTION 5 · COMPETENCY 164

Which vital center sets the basic rhythm of breathing, and where is it?

A. The cardiac center, in the pons

B. The respiratory center, in the midbrain

C. The respiratory center, in the medulla oblongata

D. The vasomotor center, in the medulla oblongata

Answer: C. The respiratory center is one of the three vital centers inside the medulla, part of the reticular formation and not visible on the surface. It sets the basic rhythm of breathing. The pons holds nuclei that adjust that rhythm, but the rhythm itself comes from the medulla.

Why the others are wrong

- **A** The cardiac center adjusts the rate and force of the heartbeat, and it is in the medulla, not the pons.
- **B** The midbrain holds the colliculi, the cerebral peduncles, and the substantia nigra. None of the three vital centers is there.
- **D** The vasomotor center is in the medulla, but its job is to adjust blood vessel diameter and so blood pressure, not breathing.

QUESTION 6 · COMPETENCY 165

The nucleus of the abducens nerve (CN VI) is housed in which region?

A. Midbrain

B. Medulla oblongata

C. Upper cervical spinal cord

D. Pons

Answer: D. The pons houses the nuclei of CN V, VI, VII, and VIII. The abducens nerve leaves at the pontomedullary junction near the midline, but its nucleus is on the pons side of that border. Use the 2-2-4-4 rule: two above the brainstem, two in the midbrain, four in the pons, four in the medulla.

Why the others are wrong

- **A** The midbrain holds the nuclei of CN III (oculomotor) and CN IV (trochlear) only.
- **B** The medulla holds the nuclei of CN IX, X, and XII, with VIII at its junction with the pons and XI grouped with it. CN VI is not in that group.
- **C** The upper cervical spinal cord holds the motor nucleus of CN XI (accessory), not CN VI.

QUESTION 7 · COMPETENCY 165

Which cranial nerve is grouped with the medulla in the 2-2-4-4 rule even though its nucleus actually sits in the upper cervical spinal cord?

A. XI Accessory

B. XII Hypoglossal

C. X Vagus

D. VIII Vestibulocochlear

Answer: A. The accessory nerve (CN XI) is counted as one of the four medullary nerves, but its motor nucleus is in the upper cervical cord. Its rootlets rise through the foramen magnum and leave the skull with the vagus nerve.

Why the others are wrong

- **B** The hypoglossal nerve (CN XII) has its nucleus in the medulla and leaves in the groove between the pyramid and the olive.
- **C** The vagus nerve (CN X) has its nucleus in the medulla and leaves in the groove behind the olive, below CN IX.
- **D** The vestibulocochlear nerve (CN VIII) has its nuclei at the junction of the pons and medulla, so it is listed with both. It is the nerve that appears beside two regions, not the one in the cervical cord.

QUESTION 8 · COMPETENCY 166

Which part of the reticular formation keeps the cerebral cortex awake and alert?

A. The descending component

B. The ascending reticular activating system

C. The substantia nigra

D. The pontine nuclei

Answer: B. The ascending reticular activating system (RAS) sends fibers up through the thalamus to the cerebral cortex. It keeps the cortex awake and alert, holds attention, and filters out unimportant sensory input. Its activity falls during sleep, and severe damage to it can cause a coma. Ascending means awake.

Why the others are wrong

- **A** The descending component sends fibers down to the spinal cord. It helps set resting muscle tone and posture and includes the medullary centers for heart rate, breathing, and blood vessel diameter.
- **C** The substantia nigra is a dark band of gray matter in the midbrain that works with the basal nuclei to help control movement. It is not part of the reticular formation.
- **D** The pontine nuclei relay motor information from the cerebral cortex to the cerebellum. They have nothing to do with keeping the cortex awake.